How-To Library

Informatica

Secure Agent Best Practices and Tuning Guidelines

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Abstract

A Secure Agent is a lightweight program that runs all tasks that you configure in Informatica Intelligent Cloud Services℠ and processes your company's data locally and securely. This article provides instructions for tuning the Secure Agent for optimal task performance in the Spring 2020 and later versions of Informatica Intelligent Cloud Services.

Supported Versions

- Informatica Intelligent Cloud Services Spring 2020

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Overview

A Secure Agent is a lightweight program that runs all tasks that you configure in Informatica Intelligent Cloud Services℠ and processes your company's data locally and securely. You might need to configure tuning properties to enable the Secure Agent to run tasks more efficiently.

You can configure tuning properties for the following types of resources:

- Linux system resources
- Amazon S3 data sources
- The Data Integration Server Secure Agent service

Additionally, you can use the RunAJob utility to run tasks concurrently.

Tuning Linux system resources

If you install the Secure Agent on a Linux machine, ensure that resources are adequately configured so that tasks can run successfully.

Check the `ulimit -a` settings on the machine and verify that they are set as follows:

```
core file size          (blocks, -c) unlimited
data seg size           (kbytes, -d) unlimited
scheduling priority             (-e) 0
file size               (blocks, -f) unlimited
pending signals                 (-i) 504323
max locked memory       (kbytes, -l) 64
max memory size         (kbytes, -m) unlimited
open files                      (-n) 64000
pipe size            (512 bytes, -p) 8
POSIX message queues     (bytes, -q) 819200
real-time priority              (-r) 0
stack size              (kbytes, -s) 10240
```

```
cpu time               (seconds, -t) unlimited
max user processes             (-u) 64000
virtual memory          (kbytes, -v) unlimited
file locks                     (-x) unlimited
```

Tuning Amazon S3 data sources

If you use tasks that read data from Amazon S3 data sources, you can tune some Amazon S3 parameters for better performance. Configure the staging folder for Amazon S3 data files and the multipart download threshold size parameters.

Staging folder for Amazon S3 data files

The reader for Amazon S3 downloads files in chunks by staging the files in a temporary staging folder. By default, the folder for staging data is `/tmp`. Ensure that adequate disk storage is available for the `/tmp` folder.

The default staging folder for Amazon S3 can also be overridden in the source object in the mapping. If there are multiple disks present on the Secure Agent machine, it is a good practice to distribute the staging folder across multiple disks. This helps reduce the disk bottleneck for Amazon S3 staging and makes staging faster, which improves performance.

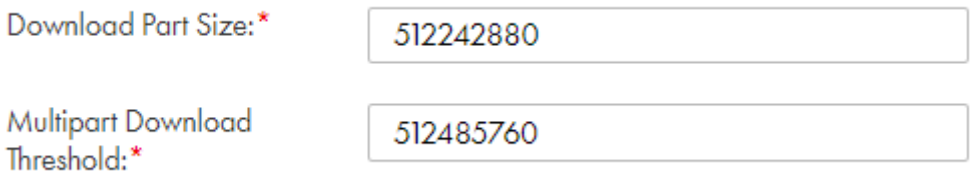
For example, if your machine has four disks /disk1, /disk2, /disk3, and /disk4, then the staging can be configured as /disk1, /disk2, /disk3, /disk4 for different Amazon S3 data sources.

Multipart download threshold size

The Amazon S3 reader downloads source files in small chunks and merges them during task execution. The default values for Download Part Size (5MB) and Multipart Download Threshold (10MB) are sometimes insufficient for processing large Amazon S3 data sources. This can cause mapping execution to fail sporadically with an error such as, "Unable to download the files from S3."

Set these parameters to a large value such as 512MB. You can override the parameter values in the advanced source properties in the Source transformation in the mapping.

The following image shows the properties:



Tuning the Data Integration Server

The Data Integration Server is the Secure Agent service that runs data integration jobs such as mapping, task, and taskflow instances. You can tune the Data Integration Server to achieve better task performance.

To improve performance, configure the maxDTMProcesses custom property and the JVM options.

You can configure these properties from the **Runtime Environments** page in Administrator. Select a Secure Agent on the **Runtime Environments** page to display and edit service properties for the agent.

For more information about configuring Secure Agent service properties, see "Configuring Secure Agent service properties" in *Secure Agent Services*.

maxDTMProcesses custom property

By default, a Secure Agent can schedule only two mapping tasks for execution. If there are more than two tasks, additional tasks are queued and then scheduled for execution when a slot becomes available. This can cause the Secure Agent machine's capacity to be underutilized.

To achieve better utilization of the CPU capacity of the Secure Agent machine and achieve a higher degree of concurrency, you can set the maxDTMProcesses custom property for the Data Integration Server to the number of parallel tasks. For example, setting this property to 16 allows 16 tasks to run simultaneously.

The following image shows configuration for the maxDTMProcesses custom property:



The recommended value for this property varies based on the connection types of the jobs that the agent runs:

- If the jobs use file-based or ODBC connections, the recommended value for the Secure Agent machine is `0.75 * # logical CPUs`. For example, if there are 24 logical CPUs on the Secure Agent machine, set this value to 18.
- If the jobs use cloud data warehouse or cloud data lake connectors, the recommended value for the Secure Agent machine is `0.33 * # logical CPUs` For example, if there are 24 logical CPUs on the Secure Agent machine, set this value to 8.

When you set the maxDTMProcesses custom property, note the following guidelines:

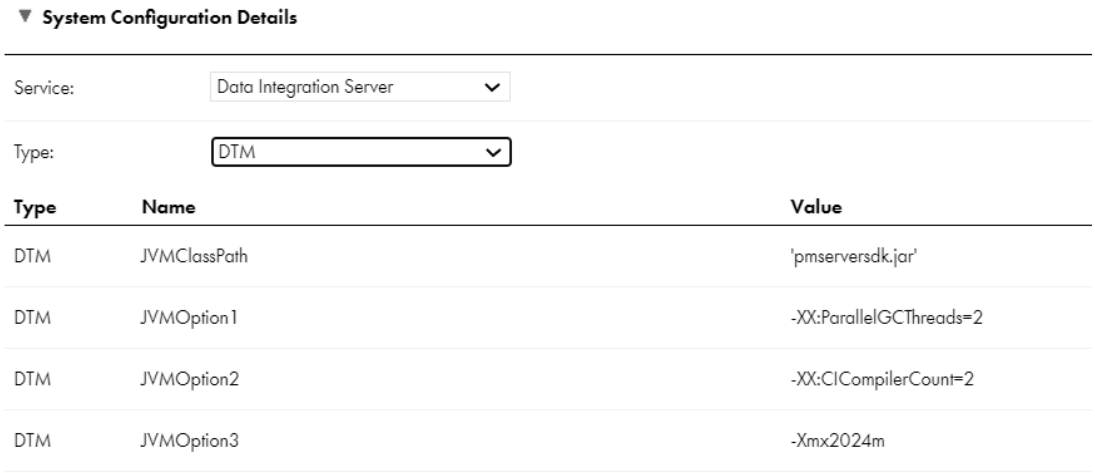
- Setting this custom property to a value higher than the number of CPUs on the Secure Agent machine can help increase parallelism for task execution, but it can also cause performance bottlenecks in task execution time.
- When you set this property, make sure that you do not exceed the terms of your license agreement. For example, if your organization is licensed for 3 Secure Agents and can run each agent on up to 4 CPUs, do not set this property to a value greater than 12.

JVM options

Each mapping task that the Data Integration Server executes runs in a unique process on the Secure Agent machine called the DTM process (pmdtm). When processing data from an Amazon Web Services S3 file system, the DTM process creates an internal Java Virtual Machine (JVM) to set up a cloud connection bridge between the Secure Agent

machine and the Amazon S3 ecosystem. There are some JVM options that you can configure for this process to achieve optimal performance.

The following image shows the JVM options that you can configure:



-XX:ParallelGCThreads

This property determines the number of garbage collection threads for the JVM. By default, it is computed as `3 + 5N/8` where `N` is the number of CPUs on the agent machine. The recommended value for this property is 2.

Tuning this property helps reduce the overall CPU utilization of the Secure Agent machine under concurrent loads by reducing unwanted context switching of the JVM threads.

-XX:CICompilerCount

This property determines the number of compiler threads for the JVM for byte code optimization. The default value is computed based on the number of CPUs of the agent machine. The recommended value for this property is 2.

Tuning this property helps reduce the overall CPU utilization of the Secure Agent machine under concurrent loads by reducing unwanted context switching of the JVM threads.

-Xmx

This property determines the heap size of the JVM for the DTM. The default value is 64MB.

For mappings that process large data volumes with multiple partitions, the default value can cause mapping failure because of insufficient Java Heap Space. The recommended value for this property is 2024MB.

Parallel task execution

You can use the RunAJob utility to run concurrent mapping tasks.

You can run individual mapping tasks using the following command:

```
./cli.sh runAJobCli -t MTT -fp $<folder path> -n $<task name> -w true &
```

The command uses the following arguments:

Argument	Description
-t	Task type. MTT is the task type for a mapping task.
-fp	Folder path to the location of the task, for example, `MyProject/Folder1`.

Argument	Description
-n	Task name.
-w	Wait flag. Determines whether to wait for the job to complete or run the job in the background.

For more information about the RunAJob utility, see "RunAJob utility" in the *REST API Reference*.

Author

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